

ABHYUDAY SINGH

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EDUCATION

The University of Texas at San Antonio

Expected May 2029

B.S. in Computer Science · GPA: 3.26

Claudia Taylor Johnson High School

May 2025

San Antonio, TX · Cumulative GPA: 3.95

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS

Libraries and Tools: Matter.js, Git, GitHub, GitHub Pages, VS Code, Microsoft Excel, Microsoft Word

PROJECTS

Physics Sandbox | [Live Demo](#) | [Code](#)

July 2026

JavaScript, Matter.js, HTML, CSS, GitHub Pages

- Built an interactive browser-based 2D physics sandbox using Matter.js with real-time rigid body simulation.
- Implemented configurable physics systems including gravity, explosions, black holes, buoyancy, and live simulation controls adjustable mid-run.
- Deployed the application with GitHub Pages and managed version control using Git and GitHub.

Pathfinder | [Live Demo](#) | [Code](#)

2026

JavaScript, HTML, CSS, GitHub Pages

- Implemented five graph search algorithms, BFS, DFS, Dijkstra, A*, and greedy best-first, with node-by-node visualization of the search frontier.
- Built a comparison mode that runs two algorithms side by side, exposing differences in nodes explored and path optimality.
- Added maze generation using recursive backtracking and Prim's algorithm, with adjustable grid resolution and step-through playback.

Singularity, Black Hole Simulator | [Live Demo](#) | [Code](#)

2026

Vanilla JavaScript, HTML5 Canvas, CSS

- Simulated hundreds of particles under Newtonian gravity with accretion disc rendering, written without external libraries.
- Built a live telemetry panel reporting frame rate, particle count, average speed, mass, and particles consumed.

EXPERIENCE

Volunteer Hour Coordinator, Hope for Triumph (Non-Profit)

Aug 2022 – Sep 2025

- Designed tracking systems for 50+ volunteers and generated activity reports used by staff.
- Applied data validation techniques to maintain accurate volunteer records across large datasets.
- Managed reporting deadlines while ensuring accuracy and consistency of records.

ACTIVITIES AND AWARDS

Hackathon Participant

Oct 2025, Jan 2026

- Built and presented working projects within 24 to 36 hour build windows across two events.

DECA (Distributive Education Clubs of America), Competitor

Aug 2023 – May 2024

- Qualified for the DECA State Competition in Automotive Marketing; competed in judged business case studies under time constraints.

President's Volunteer Service Award

Aug 2022 – May 2024

- Silver Award for 250+ volunteer hours; Bronze Award for 100+ volunteer hours.